BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: MRD-X-21(A)

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** MRD-X-21(A) VERSION: B

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e. A,B,C, or D.

Q1. The focal length of a convex mirror with a radius of curvature of 10 cm is .....

(A) + 10 cm
(B) − 10 cm
(C) − 5 cm
(D) + 5 cm

Q2. Which of the following materials will refract light more?

(A) Air
(B) Diamond
(C) Glass
(D) Water

Q3. The unit of electric potential is .....

(A) NC^{-1}
(B) Farad
(C) Coulomb
(D) Volt

Q4. Kilowatt (kWh) is the unit of electrical .....

(A) Power
(B) Energy
(C) Charge
(D) emf

Q5. Which of the following represents one ohm?

(A) VA^{-1}
(B) JS^{-1}
(C) WA^{-1}
(D) JC^{-1}

Q6. A step-up transformer increases .....

(A) Energy
(B) Power
(C) Current
(D) Voltage

Q7. The Boolean equation for the NOR function is .....

(A) $X = \bar{A}$
(B) X = AB with overline
(C) X = A+B with overline
(D) X = A

Q8. Cathode rays are the beam of .....

(A) Ions
(B) Neutrons
(C) Protons
(D) Electrons

Q9. Which one or more of the three decay processes (α, β and γ) results in a new element?

(A) Only α
(B) Only β
(C) Only γ
(D) α and β

Q10. What is the number of neutrons in the plutonium $_{92}{}^{242}Pu$

(A) 92
(B) 142
(C) 150
(D) 242

Q11. The distance travelled by wave in unit time is called .....

(A) Wave cycle
(B) Wavelength
(C) Wave speed
(D) Amplitude

Q12. The pitch of a note depends upon ..... of the sound wave.

(A) Amplitude
(B) Loudness
(C) Displacement
(D) Frequency

Section-B

Time Allowed: 2:40 Hours | **Maximum Marks:** 32

Q2. Write short answers to any eight (8) of the following parts. All parts carry equal marks.

(i) Define transverse and longitudinal waves. Give one example of each.

(ii) Why sound cannot be heard on the moon?

(iii) Define the following terms in describing the reflection of light. (a) Incident Ray (b) Reflected Ray (c) Angle of incidence (d) Angle of reflection.

(iv) How much work must be done in moving a charge of 3 C from a point at 118 V to a point at 138 V in an electric field?

(v) State the factors affecting the magnitude of induced emf.

(vi) A heating element on an electric range operating on 240 V has a resistance of 30 Ω. What current does it draw?

(vii) Can an electron at rest be set into motion with a magnetic field?

(viii) Why does a compass needle point North?

(ix) If there are four inputs in any logic gate, how many combinations are possible?

(x) Write any four uses of radioisotopes.

(xi) Which is more likely to expose, a film kept in a cardboard box, α particles or β particles? Explain.

Section-C

Maximum Marks: 21

> Note: Answer any three (3) questions. All questions carry equal marks.

Q3.

(a) [4] What is sound? How it is produced, transmitted and received?

(b) [3] What is the frequency of oscillation if the time period is 20 ms?

Q4.

(a) [4] How does the human eye work? How the defects in the eye like short-sightedness and long-sightedness is corrected by using lenses?

(b) [3] State and explain Coulomb's law? Write down its mathematical form.

Q5.

(a) [4] Differentiate between alternating current and direct current.

(b) [3] A transformer connected to a 120 V AC line is to supply 9600 V for a neon sign. What is the ratio of secondary to primary turns?

Q6.

(a) [4] Radioactive sources are said to have half-life. What is the meaning of half-life?

(b) [3] Define logic gates. Describe the operation of AND, OR logic gates by drawing their symbols and truth tables.